MIL-S-14987(MU)

17 December 1968
SUPERSEDING
PA-PD-1897
6 September 1963

MILITARY SPECIFICATION

SQUIB SWITCHES

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for two types (three classes) of squib switch used to control electrical circuits within a firing device.

1.2 Classification. This specification covers squib switches of the following types: (classes)

Type I - Delay

Class 1 - 2 normally closed and 2 normally open sets of contacts (1 second).

Class 2 - 2 normally closed and 2 normally open sets of contacts (3 seconds).

Class 3 - 2 normally closed and 2 normally open sets of contacts (8 seconds).

Type II - Non-delay

Class 1 - 2 normally closed and 2 normally open sets of contacts

Class 2 - 4 normally closed sets of contacts

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

FSC 1125

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MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

DRAWINGS

ORDNANCE CORPS

8815354 - Insulation Resistance Tester
8823371 - Switch
8823372 - Switch
8838943 - Switch, Squib, 1 Second
8838944 - Switch, Squib, 3 Second
8838945 - Switch, Squib, 8 Second
ET-8850205 - Equipment Tabulation
ET-8850267 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specification listed on the applicable drawing.

3.1.1 Piston. The piston shall be driven into the chamber by a force of 65 ± 25 pounds when tested as specified in 4.4.1.

3.2 Bridgewire Resistance.

3.2.1 Type I. The bridgewire resistance of the squib prior to assembling in the type I switch assembly and after assembly to the type I switch assembly shall be 2.2 ± 0.4 ohms when tested as specified in 4.4.2.

3.2.2 Type II. The bridgewire resistance of the squib prior to assembling in the type II switch assembly and after assembly to the type II switch assembly shall be 1.8 ± 0.4 ohms when tested as specified in 4.4.2.

3.3 Squib, prior to assembling to the type I or type II switch assembly, as applicable.

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3.3.1 No-fire energy. The squib shall have no-fire energy of 15,000 ergs at a temperature of + 160 ± 5 degrees Fahrenheit (°F). when tested as specified in 4.4.3.

3.3.2 All-fire current and functioning time.

3.3.2.1 High temperature. The squib at a temperature of + 160 ± 5°F. shall be functioned by a current of .800 ± .025 amperes and the functioning time shall comply with the applicable time specified in table I when tested as specified in 4.4.4.1.

3.3.2.2 Low temperature. The squib at a temperature of -65 ± 5°F. shall be functioned by a current of .800 ± .025 amperes and the functioning time shall comply with the applicable time specified in table I when tested as specified in 4.4.4.2.

Table I

Switch drawing No.	Initiator	Functioning time (seconds)
8823371	Squib (type II)	0.010 max.
8823372	Squib (type II)	0.010 max.
8838943	Squib (type I)	1 ± 50 percent
8838944	Squib (type I)	3 + 25, -40 percent (+165°F) 3 + 50, -0 percent (-65°F) 3 ± 40% ambient
8838945	Squib (type I)	8 + 25 percent

3.4 Assembly. The switch assembly shall comply with all requirements specified on Drawings (dwgs.) 8823371, 8823372, 8838943, 8838944, or 8838945, as applicable, and with all requirements specified in applicable specifications to the extent specified on the drawing and herein.

3.5 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.6 Contact position and resistance.

3.6.1 Prior to functioning. The position of the contacts shall be as specified in table II and the resistance of the normally closed contacts shall be .008 ohms maximum when tested as specified in 4.4.5.

Table II

Switch dwg. No.	The number of the normally closed contacts	The number of the normally open contacts
8823371	1 and 2	3 and 4
8823372	1, 2, 3 and 4	-------
8838943	1 and 4	2 and 3
8838944	1 and 4	2 and 3
8838945	1 and 4	2 and 3

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3.6.2 After functioning. The position of the contacts shall be the reverse of the position specified in table II and the resistance of the closed contacts shall be .008 ohms maximum when tested as specified in 4.4.5.

3.7 Insulation resistance.

3.7.1 Before functioning. The insulation resistance shall not be less than 50 megohms when a potential of 500 ± 50 volts direct current (v.d.c.) with current limited to 5 milliamperes (ma) is applied as specified in table III when tested as specified in 4.4.6.

Note: Refer to 3.6.1 and table II for the number of the closed and opened contacts.

Table III

(+)	(-)
1. Both terminals of normally closed contacts.	1. All other terminals and case.
2. Each terminal of normally opened contacts.	2. All other terminals and case.
3. Initiator leads (shorted together).	3. All terminals and case.

3.7.2 After functioning. The insulation resistance shall not be less than 50 megohms when a potential of 500 ± 50 v.d.c. with current limited to 5 ma is applied as specified in table IV when tested as specified in 4.4.6.

Note: Refer to 3.6.2 and table II for the number of the closed and open contacts.

Table IV

(+)	(-)
1. Both terminals of closed contacts.	1. All other terminals and case.
2. Each terminal of opened contacts.	2. All other terminals and case.
3. Each initiator lead.	3. All terminals.

3.8 Impact shock. The switch assembly shall withstand three impact shocks of 15 g's in both directions of its three mutually perpendicular planes without evidence of distortion, cracking, chipping, loosening of parts or other deterioration and after test shall comply with 3.2, 3.6.1, 3.7.1 and 3.13 when tested as specified in 4.4.7.

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3.9 Vibration. The switch assembly shall withstand vibrations through the frequency range of 5 to 500 to 5 cycles per second (c.p.s.) with a total double amplitude of 0.01 inch or an applied vibratory acceleration of 2 g's (in each direction) whichever is the lower value, without evidence of distortion, cracking, chipping, loosening of parts or other deterioration and after test shall comply with 3.2, 3.6.1, 3.7.1 and 3.13 when tested as specified in 4.4.8.

3.10 High temperature. The switch assembly shall withstand exposure to a temperature of + 165 ± 5°F. without evidence of swelling, peeling, cracking, distortion or other deterioration and during exposure shall comply with 3.2, 3.6.1, 3.7.1 and 3.13 when tested as specified in 4.4.9.

3.11 Low temperature. The switch assembly shall withstand exposure to a temperature of -65 ± 5°F. without evidence of shrinkage, embrittlement, cracks, distortion or other deterioration and during exposure shall comply with 3.2, 3.6.1, 3.7.1 and 3.13 when tested as specified in 4.4.10.

3.12 Altitude. The switch assembly shall withstand atmospheric pressure simulating a minimum altitude of 30,000 feet without evidence of distortion, cracking, chipping, loosening of parts or other deterioration and shall comply with 3.2, 3.6.1, 3.7.1 and 3.13 at a minimum altitude of 10,000 feet when tested as specified in 4.4.11.

3.13 Functioning. The switch assembly shall be functioned by a current of 0.800 ± 0.025 amperes, and the functioning time shall be in accordance with table I. After functioning, the assembly shall comply with 3.6.2, 3.7.2 and 3.14.2 when tested as specified in 4.4.12.

3.14 X-Ray.

3.14.1 Prior to functioning. The switch assembly shall be x-rayed and there shall be no evidence of any foreign material (see 6.4), nor shall any contact, slider, bridgewire, or piston be missing, misplaced or damaged when tested as specified in 4.4.13.

3.14.2 After functioning. The switch assembly shall be x-rayed and there shall be no evidence of damage to internal components (piston, slider, contacts, etc.) and the positioning of the components shall be proper when tested as specified in 4.4.13.

3.15 Workmanship.

3.15.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.15.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.15.3 Burrs. Parts shall be free from burrs.

3.15.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.15.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.15.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 15 complete assemblies (see dwg 8823371 or 8823372) as applicable, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table V. Tests shall be conducted in order specified in table V.

Table V

Order of test	Number of samples tested	Number of samples subjected to functioning after each test
1. Classification of characteristics (see 4.3.2.1)	15	-
2. Bridgewire resistance (see 3.2)	15	-
3. Contact position and resistance (see 3.6)	15	-
4. Insulation resistance (see 3.7)	15	-
5. X-Ray (3.14.1)	15	-
6. Impact shock (see 3.8)	15	3
7. Vibration (see 3.9)	12	3
8. High temperature (see 3.10)	9	3
9. Low temperature (see 3.11)	6	3
10. Altitude (see 3.12)	3	3
11. Functioning (total number) (see 3.13)	15	-
12. X-Ray (see 3.14.2)	15	-
13. Impact shock (see 3.8)	15 *	-
14. Vibration (see 3.9)	15 *	-

*Testing shall be performed on each switch as specified in 4.4.7 and 4.4.8 except that after each test only contact position and resistance (after functioning) 4.4.5 and insulation resistance (after functioning) 4.4.6 shall be performed.

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168. In addition, each lot shall contain:

a. Squibs from the same lot, samples of which have complied with 3.2 and 3.3.

b. Pistons from the same lot, samples of which have complied with 3.1.2.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table VI

Lot size	Sample size
1- 155	100% inspection
156- 200	155
201- 250	175
251- 300	190
301- 350	200
351- 400	210
401- 450	220
451- 500	225
501- 550	230
551- 600	235
601- 650	240
651- 700	243

Minor characteristic table VII

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35

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Lot size	Sample size	Lot size	Sample size
701- 750	247	86- 95	36
751- 800	250	96- 110	37
801- 900	254	111- 130	38
901-1000	258	131- 156	39
1001-1100	261	157- 190	40
1101-1200	264	191- 240	41
1201-1300	266	241- 340	42
1301-1400	268	341- 520	43
1401-1500	270	521-1120	44
1501-1600	272	1121 and over	45

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch (see dwgs 8823371 and 8823372).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for switch in accordance with applicable drawing.....	Visual
Minor:		
201.	Workmanship (see 3.15).....	Visual/Manual

4.3.2.1.2 Preparation for delivery (see section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU).....	Visual
Minor:		
201.	Workmanship (see 3.15).....	Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

	Test	Test classification
1.	Piston	Major
2.	Squib (types I and II)	
	a. Bridgewire resistance	Major
	b. No-fire energy	Major
	c. All-fire energy and functioning time	Major

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Test	Test classification
3. Switch assembly (types I and II)	
a. Bridgewire resistance	Major
b. Contact position and resistance	Major
c. Insulation resistance	Major
d. High temperature	Major
e. Low temperature	Major
f. Functioning	Major
g. X-ray	Major

Of these tests 1, 2c and 3f are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Piston (see 3.1.2). Sample pistons shall be selected from each lot and subjected to the test (see 4.4.1). Sample size and acceptance will be based on MIL-STD-105 utilizing an A.Q.L. of 1.00 percent.

4.3.3.2 Squib (type I and type II) prior to assembly in switches.

4.3.3.2.1 Bridgewire resistance (see 3.2) and no-fire energy at high temperature (see 3.3.1). A sample of 70 units shall be selected at random from the lot for these tests. Each unit shall be subjected to the bridgewire resistance test (4.4.2) and if any sample fails to comply with the requirement it shall be removed from the sample lot and replaced with an acceptable unit. Each of the samples shall be subjected to the no-fire energy test at high temperature (4.4.3.2). If any sample fails to comply with the applicable requirement, the lot represented by the sample shall not be used in the assembly of squib switches.

Note. This sample of 70 squibs shall then be subjected to the tests specified in 4.3.3.2.2, exposing the same samples to the same temperature conditions.

4.3.3.2.2 Bridgewire resistance (see 3.2) and all fire and functioning time at high and low temperature (see 3.3.2). A sample of 105 units (the 70 samples selected for the tests specified in 4.3.3.1.1 and 35 additional samples) shall be selected at random from the lot for these tests. Thirty five of the samples that were subjected to the tests specified in 4.3.3.2.1 shall be subjected to the bridgewire resistance, all-fire, and functioning time tests at high temperature (see applicable test in 4.4). The remaining 35 assemblies that were subjected to the tests specified in 4.3.3.2.1 shall be subjected to the bridgewire resistance, all-fire, and functioning time test at low temperature (see applicable paragraph in 4.4). The 35 additional samples shall be subjected to the bridgewire resistance, all-fire, and functioning time tests at low temperature (see applicable paragraph 4.4). Results of the three groups of 35 units shall be reported separately and a copy of the results shall be forwarded to Picatinny Arsenal, ATTENTION: SMUPA-NR1, Dover, New Jersey. If any sample fails to comply with the applicable requirement, the lot represented by the sample in which failure occurred shall be rejected.

4.3.3.3 Switch assembly (type I only).

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4.3.3.3.1 Bridgewire resistance (see 3.2.1); contact position and resistance (see 3.6.1) and insulation resistance (see 3.7.1) X-ray (see 3.14.1). Each switch assembly shall be subjected to these tests (see applicable paragraph in 4.4). If any assembly fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.3.3.3.2 High temperature (see 3.10), functioning at high temperature (see 3.13), contact position and resistance (see 3.6.2), insulation resistance (see 3.7.2), and X-ray (see 3.14.2). Lots of less than 100 type I (delay) switches will be accepted by functioning eight switches at each temperature, and the function times shall be within the limits specified (see table I). Lots of 100 or more shall be accepted by functioning 25 switches at each temperature, and the functioning times shall be within the limits specified (see table I). In addition, the standard deviation shall be recorded and the mean $\pm$ 3 standard deviations shall be within the limits specified (see table I).

4.3.3.3.3 High temperature (see 3.10), low temperature (see 3.11), functioning at low temperature (see 3.13), contact position and resistance (see 3.6.2), insulation resistance (see 3.7.2), and X-ray (see 3.14.2). Lots of less than 100 type I (delay) switches will be accepted by functioning eight switches at each temperature, and the function times shall be within the limits specified (see tabel I). Lots of 100 or more shall be accepted by functioning 25 switches at each temperature, and the functioning times shall be within the limits specified (see table I). In addition, the standard deviation shall be recorded and the mean $\pm$3 standard deviations shall be within the limits specified (see table I).

4.3.3.4 Switch assembly (type II only).

4.3.3.4.1 Bridgewire resistance (see 3.2.2); contact position and resistance (see 3.6.1), and insulation resistance (see 3.7.1), X-ray (see 3.14.1). Each switch assembly shall be subjected to these tests (see applicable paragraph in 4.4). If any assembly fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.3.3.4.2 High temperature (see 3.10), functioning at high temperature (see 3.13), contact position and resistance (see 3.6.2), insulation resistance (see 3.7.2) and X-ray (see 3.14.2). One sample switch assembly shall be selected from each 25 assemblies or portion thereof produced and subjected to these tests (see applicable paragraph in 4.4). If any assembly fails to comply with the applicable requirement, the lot represented shall be rejected.

4.3.3.4.3 High temperature (see 3.10), low temperature (see 3.11), functioning at low temperature (see 3.13), contact position and resistance (see 3.6.2), insulation resistance (see 3.7.2) and X-ray (see 3.14.2). Sample switch assemblies shall be selected in accordance with table VIII and subjected to these tests (see applicable paragraph in 4.4). If any assembly fails to comply with the applicable requirement, the lot represented by the sample shall be rejected.

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Table VIII

Lot size	Sample size
2-8	1
9-15	2
16-25	3
26-40	4
41-65	8
66-110	13
111-180	22
181-300	36
300 and over	MIL-STD-105 (A.Q.L. 0.25 percent)

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8850205 and 8850267. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Piston (see 3.1.1). This test shall be performed by applying a force until the piston passes through the cavity of the cylinder chamber. An indicator shall be used to record the amount of force used.

4.4.2 Bridgewire resistance (see 3.2). The bridgewire resistance shall be determined utilizing method 307 of Standard MIL-STD-202. The shorting clip shall be removed and the measurement taken across the leads. The current shall be limited to 5 milliamperes.

4.4.3 No-fire energy (see 3.3.1). The squib shall be placed in a test chamber and the temperature shall be raised to +165°F. After stabilization, (see 6.4) the assembly shall be exposed to this temperature for a minimum of 2 hours. While still at this temperature, the assembly shall be immediately tested in accordance with 4.4.2 and then subjected to a discharge from a 3.83 ± 1 percent microfarad capacitor charged to 28 volts and the squib shall not function.

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4.4.4 All-fire current and functioning time (see 3.3.2).

4.4.4.1 High temperature (see 3.3.2.1). The squib shall be placed in a test chamber and the temperature shall be raised to +165°F. After stabilization (see 6.11), the assembly shall be exposed to this temperature for a minimum of 2 hours. While still at this temperature, the assembly shall be immediately tested in accordance with 4.4.2. The squib shall then be subjected to the voltage from a d.c. source capable of supplying a current of .800 ± .025 amperes and the squib shall function within the specified time. The potential shall be limited to 18 volts. Functioning time shall start at the application of the voltage and end at the functioning of the initiator. After functioning, the insulation resistance shall be determined in accordance with 4.4.6 for compliance with table IV, as applicable.

4.4.4.2 Low temperature (see 3.3.2.2). The squib shall be tested as specified in 4.4.4.1. except that the temperature shall be lowered to -65°F.

4.4.5 Contact position and resistance (see 3.6). The contact position and the resistance of the closed contacts shall be determined utilizing method 307 of Standard MIL-STD-202.

4.4.6 Insulation resistance (see 3.7). The insulation resistance shall be determined utilizing the insulation resistance tester specified on test panel dwg. 8815354. The potential shall be applied until the meter has ceased fluctuating and a rising or steady value has been obtained. However, the potential shall not be applied for less than 5 seconds. The current shall be limited to not more than 5 milliamperes as a safety measure.

4.4.7 Impact shock (see 3.8). The switch assembly shall be tested as specified in the shock test, procedure V of Specification MIL-E-5272 except that the assembly shall not be operated and the crash-safety portion shall not apply. The switch assembly shall be subjected to a total of 18 shocks. After completion of the shock test, the assembly shall be tested as specified in 4.4.2, 4.4.5, 4.4.6 and 4.4.12.

Note: Applicable to first article samples subjected to this test after being functioned; after completion of the shock test, the assembly shall be tested as specified in 4.4.5 and 4.4.6 and the resistance of the closed contacts shall be 0.100 maximum.

4.4.8 Vibration (see 3.9). The switch assembly shall be vibrated through the frequency range in 15 minutes. The total time of vibration shall be a minimum of 45 minutes in each of the three mutually perpendicular planes. Resonant frequencies shall be noted and the assembly shall be vibrated, at each resonant frequency in the same plane that resonance was noted, for 15 minutes. Upon completion of vibration testing, the assembly shall be tested in accordance with 4.4.2, 4.4.5, 4.4.6 and 4.4.12.

Note: Applicable to initial production samples subjected to this test after being functioned; after completion of the vibration test, the assembly shall be tested as specified in 4.4.5 and 4.4.6 and the resistance of the closed contacts shall be 0.100 maximum.

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4.4.9 High temperature (see 3.10). The switch assembly shall be placed in a test chamber and the temperature within the chamber raised to +165 ± 5°F. After stabilization (see 6.4), the assembly shall be exposed to this temperature for a minimum of 4 hours. While still at this temperature the assembly shall be tested as specified in 4.4.2, 4.4.5, 4.4.6 and 4.4.12.

4.4.10 Low temperature (see 3.11). The switch assembly shall be placed in a test chamber and the temperature within the chamber lowered to -65 ±5°F. After stabilization (see 6.4), the assembly shall be exposed to this temperature for a minimum of 4 hours. While still at this temperature the assembly shall be tested as specified in 4.4.2, 4.4.5, 4.4.6 and 4.4.12.

4.4.11 Altitude (see 3.12). The switch shall be placed in an altitude chamber and the internal pressure shall be reduced until a pressure equivalent to a minimum altitude of 30,000 feet is obtained. The temperature within the chamber shall be -65 ±5°F. After stabilization (see 6.4), the assembly shall be subjected to the specified temperature and pressure for a minimum of 1 hour. The pressure shall then be slowly and uniformly reduced to a pressure equivalent to a minimum altitude of 10,000 feet, but the temperature shall be maintained at -65 ±5°F. At this simulated altitude of 10,000 feet, minimum, the assembly shall be tested as specified in 4.4.2, 4.4.5, 4.4.6 and 4.4.12.

4.4.12 Functioning (see 3.13). The switch assembly shall function when subjected to the voltage from a direct current (d.c.), source capable of supplying .800 ± .025 amperes. The potential shall be limited to 18 volts. (Note: Separation or cracking of the sealing adhesive upon functioning shall not be cause for rejection provided other requirements are met). Functioning time shall be measured from the application of the voltage to the firing circuit until all of the individual switch contacts have operated. Immediately after functioning the assembly shall be tested as specified in 4.4.5, 4.4.6 and 4.4.13.

4.4.13 X-ray (see 3.14). The switch assembly shall be x-rayed from the top and side views. The negatives produced shall clearly indicate compliance with the requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The squib switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The squib switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The squib switch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Temperature stabilization. The temperature is considered stabilized when the peaks (both highest and lowest) continually observed over 5 minutes, minimum do not exceed the nominal temperature plus the tolerance specified.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1125-A003

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